

```

Internet#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Internet#
Internet#
Internet#ping www.google.com

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Internet#
Internet#
Internet#ping google.com

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
Internet#

```

Figure A.2: Screenshot 1.1 shows Pinging Internet Router to 8.8.8.8 ([www.google.com](http://www.google.com))

```

Internet#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          16.4.1.1        YES NVRAM  up          up
FastEthernet0/1          16.4.1.5        YES NVRAM  up          up
Loopback0                 8.8.8.8         YES NVRAM  up          up
Internet#
Internet#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    16.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       16.4.1.4/30 is directly connected, FastEthernet0/1
S       16.4.0.0/16 [1/0] via 16.4.1.2
C       16.4.1.0/30 is directly connected, FastEthernet0/0
    8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8 is directly connected, Loopback0
S*     0.0.0.0/0 is directly connected, Null0
Internet#

```

Figure A.3: Screenshot 1.2 shows Internet Router - Interface Status & Routing Table

```

ISP1#ping 16.4.1.1 ! Ping Internet Router

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/14/24 ms
ISP1#
ISP1#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/20/28 ms
ISP1#

```

Figure A.4: Screenshot 1.3 shows ISP1 – Ping Internet Router

```

ISP2#ping 16.4.1.1 ! Ping Internet Router

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/59/92 ms
ISP2#
ISP2#
ISP2#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/22/44 ms
ISP2#

```

Figure A.5: Screenshot 1.4 shows ISP2 – Ping Internet Router

```

R1#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/33/44 ms
R1#
R1#ping 16.4.1.51 ! Ping R2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.51, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/14/20 ms
R1#
R1#ping 16.4.1.50 ! Ping GLBP Virtual IP

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#

```

Figure A.6: Screenshot 1.5 shows R1 – Ping Internet, R2 & GLBP IP Address

```
Dist_Fiber_01#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/125/164 ms
Dist_Fiber_01#
Dist_Fiber_01#ping 16.4.1.18 ! Ping R1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.18, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/61/92 ms
Dist_Fiber_01#
Dist_Fiber_01#ping 16.4.1.14 ! Ping R2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.14, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/60/68 ms
Dist_Fiber_01#
```

Figure A.7: Screenshot of Distribution\_Fiber01 – Ping Internet, R1 & R2

```
DistributionFiber02#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 96/111/124 ms
DistributionFiber02#
DistributionFiber02#ping 16.4.1.18 ! Ping R1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.18, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/20/24 ms
DistributionFiber02#
DistributionFiber02#ping 16.4.1.14 ! Ping R2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.1.14, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/18/48 ms
DistributionFiber02#
DistributionFiber02#ping 16.4.2.5 ! Ping CORE1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.2.5, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/20/40 ms
DistributionFiber02#
```

Figure A.8: Screenshot of Distribution\_Fiber02 – Ping Internet, R1, R2 & CORE1



```

CORE1#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/113/136 ms
CORE1#
CORE1#ping 16.4.2.2 ! Ping CORE2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/15/28 ms
CORE1#
CORE1#ping 16.4.2.10 ! Ping DistFiber01

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.2.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/19/24 ms
CORE1#

```

Figure A.9: Screenshot of CORE1 – Ping Internet, CORE2, DistributionFiber01

```

CORE2#ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/107/128 ms
CORE2#
CORE2#ping 16.4.2.1 ! Ping CORE1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/53/68 ms
CORE2#
CORE2#ping 16.4.2.10 ! Ping DistFiber02

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 16.4.2.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/20/44 ms
CORE2#
CORE2#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

```

| H | Address   | Interface | Hold (sec) | Uptime   | SRTT (ms) | RTO | Q Cnt | Seq Num |
|---|-----------|-----------|------------|----------|-----------|-----|-------|---------|
| 3 | 16.4.2.18 | Fa1/1     | 12         | 00:31:25 | 113       | 678 | 0     | 133     |
| 2 | 16.4.2.22 | Fa1/0     | 12         | 00:31:26 | 111       | 666 | 0     | 117     |
| 1 | 16.4.2.1  | Fa0/0     | 11         | 00:31:29 | 78        | 468 | 0     | 132     |
| 0 | 16.4.2.5  | Fa0/1     | 12         | 00:31:29 | 67        | 402 | 0     | 131     |

```

CORE2#

```

Figure A.10: Screenshot of CORE2 – Ping Internet, CORE1 & DistributionFiber02

```

SA-PC1102> ping 16.3.100.11

16.3.100.11 icmp_seq=1 timeout
84 bytes from 16.3.100.11 icmp_seq=2 ttl=61 time=43.578 ms
84 bytes from 16.3.100.11 icmp_seq=3 ttl=61 time=61.492 ms
84 bytes from 16.3.100.11 icmp_seq=4 ttl=61 time=50.768 ms
84 bytes from 16.3.100.11 icmp_seq=5 ttl=61 time=92.138 ms

SA-PC1102>
SA-PC1102>
SA-PC1102> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=249 time=88.616 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=249 time=89.879 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=249 time=89.739 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=249 time=94.530 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=249 time=99.046 ms

SA-PC1102> █

```

Figure A.11: Screenshot of SA-PC1102 (VLAN311) – Ping Internet & SA-PC6301 (VLAN331)

```

SA-PC3201> ping 8.8.8.8

8.8.8.8 icmp_seq=1 timeout
84 bytes from 8.8.8.8 icmp_seq=2 ttl=249 time=112.679 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=249 time=71.500 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=249 time=82.905 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=249 time=70.157 ms

SA-PC3201> ping 16.1.82.1

84 bytes from 16.1.82.1 icmp_seq=1 ttl=253 time=55.263 ms
84 bytes from 16.1.82.1 icmp_seq=2 ttl=253 time=47.557 ms
84 bytes from 16.1.82.1 icmp_seq=3 ttl=253 time=39.085 ms
84 bytes from 16.1.82.1 icmp_seq=4 ttl=253 time=69.779 ms
84 bytes from 16.1.82.1 icmp_seq=5 ttl=253 time=29.565 ms

SA-PC3201> ping 16.1.82.10

84 bytes from 16.1.82.10 icmp_seq=1 ttl=61 time=52.168 ms
84 bytes from 16.1.82.10 icmp_seq=2 ttl=61 time=37.528 ms
84 bytes from 16.1.82.10 icmp_seq=3 ttl=61 time=58.684 ms
84 bytes from 16.1.82.10 icmp_seq=4 ttl=61 time=54.255 ms
84 bytes from 16.1.82.10 icmp_seq=5 ttl=61 time=33.228 ms

SA-PC3201> show

NAME      IP/MASK      GATEWAY      MAC      LPORT  RHOST:PORT
SA-PC3216.1.90.10/24      16.1.90.1      00:50:79:66:68:01  20824  127.0.0.1:20825
          fe80::250:79ff:fe66:6801/64

```

Figure A.12: Screenshot of SA-PC3201 (VLAN331) – Ping Internet, Building 101 Gateway, SA-PC1101 (VLAN311)

```

NAME      IP/MASK      GATEWAY      MAC      LPORT  RHOST:PORT
SA-PC2S16.3.63.84/26      16.3.63.65      00:50:79:66:68:0c      20830      127.0.0.1:20831
      fe80::250:79ff:fe66:680c/64

SA-PC2SB01> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=249 time=105.160 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=249 time=83.518 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=249 time=147.272 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=249 time=100.989 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=249 time=86.003 ms

SA-PC2SB01> ping 16.3.63.20

84 bytes from 16.3.63.20 icmp_seq=1 ttl=63 time=32.300 ms
84 bytes from 16.3.63.20 icmp_seq=2 ttl=63 time=44.463 ms
84 bytes from 16.3.63.20 icmp_seq=3 ttl=63 time=12.692 ms
84 bytes from 16.3.63.20 icmp_seq=4 ttl=63 time=21.757 ms
84 bytes from 16.3.63.20 icmp_seq=5 ttl=63 time=16.411 ms

SA-PC2SB01> █

```

Figure A.13: Screenshot of SA-PC2SB01 (VLAN321) – Ping Internet & SA-PC1B01 (VLAN311)

The screenshot shows the configuration window for SA-SWSB0102. The General tab is active, showing the Name as SA-SWSB0102 and Console type as none. The Settings tab is also visible, showing Port 8, VLAN 999, Type access, and QinQ EtherType 0x8100. The Ports table lists 8 ports with their respective VLANs and types.

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Figure A.14: SA-SWSB0102 Switch Configuration (Layer 2)

**SA-SW10111 configuration**

General

Name: SA-SW10111

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 1    | access |           |
| 5    | 1    | access |           |
| 6    | 1    | access |           |
| 7    | 1    | access |           |

Buttons: Add, Delete, Reset, OK, Cancel, Apply

Figure A.15: SA-SW10111 Switch Configuration (Layer 2)

**SA-SW10112 configuration**

General

Name: SA-SW10112

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 1    | access |           |
| 5    | 1    | access |           |
| 6    | 1    | access |           |
| 7    | 1    | access |           |

Buttons: Add, Delete, Reset, OK, Cancel, Apply

Figure A.16: SA-SW10112 Switch Configuration (Layer 2)



Node properties

### SA-SW10211 configuration

General

Name: SA-SW10211

Console type: none

Settings

Port: 8

VLAN: 331

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 1    | access |           |
| 5    | 1    | access |           |
| 6    | 1    | access |           |
| 7    | 1    | access |           |

Buttons: Add, Delete, Reset, OK, Cancel, Apply

Figure A.17: SA-SW10211 Switch Configuration (Layer 2)

Node properties

### SA-SW10212 configuration

General

Name: SA-SW10212

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 1    | access |           |
| 5    | 1    | access |           |
| 6    | 1    | access |           |
| 7    | 1    | access |           |

Buttons: Add, Delete, Reset, OK, Cancel, Apply

Figure A.18: SA-SW10212 Switch Configuration (Layer 2)



Node properties

## SA-SW20111 configuration

General

Name: SA-SW20111

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 1    | access |           |
| 5    | 1    | access |           |
| 6    | 1    | access |           |
| 7    | 1    | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.19: SA-SW20111 Switch Configuration (Layer 2)

Node properties

## SA-SW20112 configuration

General

Name: SA-SW20112

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 1    | access |           |
| 5    | 1    | access |           |
| 6    | 1    | access |           |
| 7    | 1    | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.20: SA-SW20112 Switch Configuration (Layer 2)

Node properties

## SA-SW20211 configuration

General

Name: SA-SW20211

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.21: SA-SW20211 Switch Configuration (Layer 2)

Node properties

## SA-SW20212 configuration

General

Name: SA-SW20212

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.22: SA-SW20212 Switch Configuration (Layer 2)



Node properties

## SA-SW30111 configuration

General

Name: SA-SW30111

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.23: SA-SW30111 Switch Configuration (Layer 2)

Node properties

## SA-SW30112 configuration

General

Name: SA-SW30112

Console type: none

Settings

Port: 8

VLAN: 999

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Buttons: Add, Delete, Reset, OK, Cancel, Apply

Figure A.24: SA-SW30112 Switch Configuration (Layer 2)

Node properties

### SA-SW30211 configuration

General

Name: SA-SW30211

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.25: SA-SW30111 Switch Configuration (Layer 2)

Node properties

## SA-SW30212 configuration

General

Name: SA-SW30212

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Ports

| Port | VLAN | Type   | EtherType |
|------|------|--------|-----------|
| 0    | 1    | dot1q  |           |
| 1    | 311  | access |           |
| 2    | 321  | access |           |
| 3    | 331  | access |           |
| 4    | 999  | access |           |
| 5    | 999  | access |           |
| 6    | 999  | access |           |
| 7    | 999  | access |           |

Add Delete

Reset OK Cancel Apply

Figure A.26: SA-SW20212 Switch Configuration (Layer 2)

## IP Addressing and Networks

IP Addresses on interfaces for routers facing other routers.

| Name            | Interface | IPV4 Address | Subnet Mask     | Network Address | Broadcast Address | Gateway         |
|-----------------|-----------|--------------|-----------------|-----------------|-------------------|-----------------|
| <b>Internet</b> | –         | –            | –               | –               | –                 | <b>N/A</b>      |
| Internet        | Loopback0 | 8.8.8.8      | –               | –               | –                 | –               |
| Internet        | F0/0      | 16.4.1.1     | 255.255.255.252 | 16.4.1.0        | 16.4.1.3          | → ISP1          |
| Internet        | F0/1      | 16.4.1.5     | 255.255.255.252 | 16.4.1.4        | 16.4.1.7          | → ISP2          |
| <b>ISP1</b>     | –         | –            | –               | –               | –                 | <b>16.4.1.1</b> |
| ISP1            | F0/0      | 16.4.1.2     | 255.255.255.252 | 16.4.1.0        | 16.4.1.3          | → Internet      |
| ISP1            | F0/1      | 16.4.1.9     | 255.255.255.252 | 16.4.1.8        | 16.4.1.11         | → R1            |
| ISP1            | F1/0      | 16.4.1.13    | 255.255.255.252 | 16.4.1.12       | 16.4.1.15         | → R2            |
| <b>ISP2</b>     | –         | –            | –               | –               | –                 | <b>16.4.1.5</b> |
| ISP2            | F0/0      | 16.4.1.6     | 255.255.255.252 | 16.4.1.4        | 16.4.1.7          | → Internet      |
| ISP2            | F1/0      | 16.4.1.17    | 255.255.255.252 | 16.4.1.16       | 16.4.1.19         | → R1            |
| ISP2            | F0/1      | 16.4.1.21    | 255.255.255.252 | 16.4.1.20       | 16.4.1.23         | → R2            |

Table 1.1: WAN and Core Router Interface IP Addressing



| Name      | Interface | IPv4 Address                   | Subnet Mask     | Network Address | Broadcast Address | Gateway       |
|-----------|-----------|--------------------------------|-----------------|-----------------|-------------------|---------------|
| <b>R1</b> | –         | –                              | –               | –               | –                 | <b>N/A</b>    |
| R1        | F0/1      | 16.4.1.10                      | 255.255.255.252 | 16.4.1.8        | 16.4.1.11         | → ISP1        |
| R1        | F1/0      | 16.4.1.18                      | 255.255.255.252 | 16.4.1.16       | 16.4.1.19         | → ISP2        |
| R1        | F1/1      | 16.4.1.49,<br>16.4.1.50 (GLBP) | 255.255.255.248 | 16.4.1.48       | 16.4.1.55         | → R2 (.51)    |
| R1        | F0/0      | 16.4.1.29                      | 255.255.255.252 | 16.4.1.28       | 16.4.1.31         | → DistFiber01 |
| R1        | F1/2      | 16.4.1.33                      | 255.255.255.252 | 16.4.1.32       | 16.4.1.35         | → DistFiber02 |
| <b>R2</b> | –         | –                              | –               | –               | –                 | <b>N/A</b>    |
| R2        | F1/0      | 16.4.1.14                      | 255.255.255.252 | 16.4.1.12       | 16.4.1.15         | → ISP2        |
| R2        | F0/1      | 16.4.1.22                      | 255.255.255.252 | 16.4.1.20       | 16.4.1.23         | → ISP1        |
| R2        | F1/1      | 16.4.1.51,<br>16.4.1.50 (GLBP) | 255.255.255.248 | 16.4.1.48       | 16.4.1.55         | → R1 (.49)    |
| R2        | F1/2      | 16.4.1.37                      | 255.255.255.252 | 16.4.1.36       | 16.4.1.39         | → DistFiber01 |
| R2        | F0/0      | 16.4.1.41                      | 255.255.255.252 | 16.4.1.40       | 16.4.1.43         | → DistFiber02 |

Table 1.2: WAN and Core Router Interface IP Addressing



| Name         | Interface | IPV4 Address | Subnet Mask     | Network Address | Broadcast Address | Gateway       |
|--------------|-----------|--------------|-----------------|-----------------|-------------------|---------------|
| <b>CORE1</b> | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| CORE1        | F0/0      | 16.4.2.1     | 255.255.255.252 | 16.4.2.0        | 16.4.2.3          | → CORE2       |
| CORE1        | F0/1      | 16.4.2.5     | 255.255.255.252 | 16.4.2.4        | 16.4.2.7          | → CORE2       |
| CORE1        | F1/0      | 16.4.2.9     | 255.255.255.252 | 16.4.2.8        | 16.4.2.11         | → DistFiber01 |
| CORE1        | F1/1      | 16.4.2.13    | 255.255.255.252 | 16.4.2.12       | 16.4.2.15         | → DistFiber02 |
| CORE1        | F1/10     | 16.4.2.25    | 255.255.255.252 | 16.4.2.24       | 16.4.2.27         | → DistFiber03 |
| CORE1        | F1/11     | 16.4.2.29    | 255.255.255.252 | 16.4.2.28       | 16.4.2.31         | → DistFiber03 |
| CORE1        | F1/12     | 16.4.2.33    | 255.255.255.252 | 16.4.2.32       | 16.4.2.35         | → DistFiber04 |
| CORE1        | F1/13     | 16.4.2.37    | 255.255.255.252 | 16.4.2.36       | 16.4.2.39         | → DistFiber04 |
| CORE1        | F1/2      | 16.4.5.2     | 255.255.255.252 | 16.4.5.0        | 16.4.5.3          | → DistFiber05 |
| CORE1        | F1/3      | 16.4.5.10    | 255.255.255.252 | 16.4.5.8        | 16.4.5.11         | → DistFiber06 |
| <b>CORE2</b> | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| CORE2        | F0/0      | 16.4.2.2     | 255.255.255.252 | 16.4.2.0        | 16.4.2.3          | → CORE1       |
| CORE2        | F0/1      | 16.4.2.6     | 255.255.255.252 | 16.4.2.4        | 16.4.2.7          | → CORE1       |
| CORE2        | F1/1      | 16.4.2.17    | 255.255.255.252 | 16.4.2.16       | 16.4.2.19         | → DistFiber01 |

|       |       |           |                 |           |           |               |
|-------|-------|-----------|-----------------|-----------|-----------|---------------|
| CORE2 | F1/0  | 16.4.2.21 | 255.255.255.252 | 16.4.2.20 | 16.4.2.23 | → DistFiber02 |
| CORE2 | F1/12 | 16.4.2.41 | 255.255.255.252 | 16.4.2.40 | 16.4.2.43 | → DistFiber03 |
| CORE2 | F1/13 | 16.4.2.45 | 255.255.255.252 | 16.4.2.44 | 16.4.2.47 | → DistFiber03 |
| CORE2 | F1/10 | 16.4.2.49 | 255.255.255.252 | 16.4.2.48 | 16.4.2.51 | → DistFiber04 |
| CORE2 | F1/11 | 16.4.2.53 | 255.255.255.252 | 16.4.2.52 | 16.4.2.55 | → DistFiber04 |
| CORE2 | F1/3  | 16.4.5.6  | 255.255.255.252 | 16.4.5.4  | 16.4.5.7  | → DistFiber05 |
| CORE2 | F1/4  | 16.4.5.14 | 255.255.255.252 | 16.4.5.12 | 16.4.5.15 | → DistFiber06 |

Table 1.3: WAN and Core Router Interface IP Addressing

| Name                       | Interface | IPv4 Address | Subnet Mask     | Network Address | Broadcast Address | Gateway    |
|----------------------------|-----------|--------------|-----------------|-----------------|-------------------|------------|
| <b>DistributionFiber01</b> | –         | –            | –               | –               | –                 | <b>N/A</b> |
| DistributionFiber01        | F0/0      | 16.4.1.30    | 255.255.255.252 | 16.4.1.28       | 16.4.1.31         | → R2       |
| DistributionFiber01        | F0/1      | 16.4.1.38    | 255.255.255.252 | 16.4.1.36       | 16.4.1.39         | → R1       |
| DistributionFiber01        | F1/0      | 16.4.2.10    | 255.255.255.252 | 16.4.2.8        | 16.4.2.11         | → CORE2    |
| DistributionFiber01        | F1/1      | 16.4.2.18    | 255.255.255.252 | 16.4.2.16       | 16.4.2.19         | → CORE1    |
| <b>DistributionFiber02</b> | –         | –            | –               | –               | –                 | <b>N/A</b> |
| DistributionFiber02        | F0/1      | 16.4.1.34    | 255.255.255.252 | 16.4.1.32       | 16.4.1.35         | → R1       |
| DistributionFiber02        | F0/0      | 16.4.1.42    | 255.255.255.252 | 16.4.1.40       | 16.4.1.43         | → R2       |
| DistributionFiber02        | F1/1      | 16.4.2.14    | 255.255.255.252 | 16.4.2.12       | 16.4.2.15         | → CORE1    |

|                            |       |           |                 |           |           |                             |
|----------------------------|-------|-----------|-----------------|-----------|-----------|-----------------------------|
| DistributionFiber02        | F1/0  | 16.4.2.22 | 255.255.255.252 | 16.4.2.20 | 16.4.2.23 | → CORE2                     |
| <b>DistributionFiber03</b> | —     | —         | —               | —         | —         | <b>N/A</b>                  |
| DistributionFiber03        | F1/0  | 16.4.6.1  | 255.255.255.252 | 16.4.6.0  | 16.4.6.3  | → SA-Edge-SW20211<br>(SW13) |
| DistributionFiber03        | F1/1  | 16.4.6.5  | 255.255.255.252 | 16.4.6.4  | 16.4.6.7  | → SA-Edge-SW10211<br>(SW6)  |
| DistributionFiber03        | F1/2  | 16.4.6.9  | 255.255.255.252 | 16.4.6.8  | 16.4.6.11 | → SA-Edge-SW10111<br>(SW2)  |
| DistributionFiber03        | F1/3  | 16.4.6.13 | 255.255.255.252 | 16.4.6.12 | 16.4.6.15 | → SA-Edge-SW20111<br>(SW9)  |
| DistributionFiber03        | F1/4  | 16.4.6.17 | 255.255.255.252 | 16.4.6.16 | 16.4.6.19 | → SA-Edge-SW30111           |
| DistributionFiber03        | F1/5  | 16.4.6.21 | 255.255.255.252 | 16.4.6.20 | 16.4.6.23 | → SA-Edge-SW30211<br>(SW10) |
| DistributionFiber03        | F1/10 | 16.4.2.26 | 255.255.255.252 | 16.4.2.24 | 16.4.2.27 | → CORE1                     |
| DistributionFiber03        | F1/11 | 16.4.2.30 | 255.255.255.252 | 16.4.2.28 | 16.4.2.31 | → CORE1                     |
| DistributionFiber03        | F1/12 | 16.4.2.42 | 255.255.255.252 | 16.4.2.40 | 16.4.2.43 | → CORE2                     |
| DistributionFiber03        | F1/13 | 16.4.2.46 | 255.255.255.252 | 16.4.2.44 | 16.4.2.47 | → CORE2                     |
| <b>DistributionFiber04</b> | —     | —         | —               | —         | —         | <b>N/A</b>                  |
| DistributionFiber04        | F1/0  | 16.4.7.1  | 255.255.255.252 | 16.4.7.0  | 16.4.7.3  | → SA-Edge-SW20211<br>(SW13) |
| DistributionFiber04        | F1/1  | 16.4.7.5  | 255.255.255.252 | 16.4.7.4  | 16.4.7.7  | → SA-Edge-SW10211<br>(SW6)  |

|                            |       |           |                 |           |           |                             |
|----------------------------|-------|-----------|-----------------|-----------|-----------|-----------------------------|
| DistributionFiber04        | F1/2  | 16.4.7.9  | 255.255.255.252 | 16.4.7.8  | 16.4.7.11 | → SA-Edge-SW10111<br>(SW2)  |
| DistributionFiber04        | F1/3  | 16.4.7.13 | 255.255.255.252 | 16.4.7.12 | 16.4.7.15 | → SA-Edge-SW20111<br>(SW9)  |
| DistributionFiber04        | F1/4  | 16.4.7.17 | 255.255.255.252 | 16.4.7.16 | 16.4.7.19 | → SA-Edge-SW30111           |
| DistributionFiber04        | F1/5  | 16.4.7.21 | 255.255.255.252 | 16.4.7.20 | 16.4.7.23 | → SA-Edge-SW30211<br>(SW10) |
| DistributionFiber04        | F1/10 | 16.4.2.50 | 255.255.255.252 | 16.4.2.48 | 16.4.2.51 | → CORE2                     |
| DistributionFiber04        | F1/11 | 16.4.2.54 | 255.255.255.252 | 16.4.2.52 | 16.4.2.55 | → CORE2                     |
| DistributionFiber04        | F1/12 | 16.4.2.34 | 255.255.255.252 | 16.4.2.32 | 16.4.2.35 | → CORE1                     |
| DistributionFiber04        | F1/13 | 16.4.2.38 | 255.255.255.252 | 16.4.2.36 | 16.4.2.39 | → CORE1                     |
| <b>DistributionFiber05</b> | —     | —         | —               | —         | —         | <b>N/A</b>                  |
| DistributionFiber05        | F0/0  | 16.4.3.1  | 255.255.255.252 | 16.4.3.0  | 16.4.3.3  | → SA-ToR-SWSB01             |
| DistributionFiber05        | F0/1  | 16.4.3.5  | 255.255.255.252 | 16.4.3.4  | 16.4.3.7  | → SA-ToR-SWSB02             |
| DistributionFiber05        | F1/2  | 16.4.5.1  | 255.255.255.252 | 16.4.5.0  | 16.4.5.3  | → CORE1                     |
| DistributionFiber05        | F1/3  | 16.4.5.5  | 255.255.255.252 | 16.4.5.4  | 16.4.5.7  | → CORE2                     |
| <b>DistributionFiber06</b> | —     | —         | —               | —         | —         | <b>N/A</b>                  |
| DistributionFiber06        | F1/0  | 16.4.4.1  | 255.255.255.252 | 16.4.4.0  | 16.4.4.3  | → SA-ToR-SWSB01             |
| DistributionFiber06        | F1/5  | 16.4.4.5  | 255.255.255.252 | 16.4.4.4  | 16.4.4.7  | → SA-ToR-SWSB02             |
| DistributionFiber06        | F1/3  | 16.4.5.9  | 255.255.255.252 | 16.4.5.8  | 16.4.5.11 | → CORE1                     |
| DistributionFiber06        | F1/4  | 16.4.5.13 | 255.255.255.252 | 16.4.5.12 | 16.4.5.15 | → CORE2                     |



Table 2: Distribution Layer Switch Interface IP Addressing

| Name                   | Interface | IPV4 Address | Subnet Mask     | Network Address | Broadcast Address | Gateway       |
|------------------------|-----------|--------------|-----------------|-----------------|-------------------|---------------|
| <b>SA-ToR-SWSB01</b>   | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| SA-ToR-SWSB01          | F0/0      | 16.4.3.2     | 255.255.255.252 | 16.4.3.0        | 16.4.3.3          | → DistFiber05 |
| SA-ToR-SWSB01          | F1/0      | 16.4.4.2     | 255.255.255.252 | 16.4.4.0        | 16.4.4.3          | → DistFiber06 |
| <b>SA-ToR-SWSB02</b>   | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| SA-ToR-SWSB02          | F0/1      | 16.4.3.6     | 255.255.255.252 | 16.4.3.4        | 16.4.3.7          | → DistFiber05 |
| SA-ToR-SWSB02          | F1/5      | 16.4.4.6     | 255.255.255.252 | 16.4.4.4        | 16.4.4.7          | → DistFiber06 |
| <b>SA-Edge-SW10111</b> | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| SA-Edge-SW10111        | F0/0      | 16.4.6.10    | 255.255.255.252 | 16.4.6.8        | 16.4.6.11         | → DistFiber03 |
| SA-Edge-SW10111        | F1/0      | 16.4.7.10    | 255.255.255.252 | 16.4.7.8        | 16.4.7.11         | → DistFiber04 |
| <b>SA-Edge-SW10211</b> | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| SA-Edge-SW10211        | F0/0      | 16.4.6.6     | 255.255.255.252 | 16.4.6.4        | 16.4.6.7          | → DistFiber03 |
| SA-Edge-SW10211        | F1/0      | 16.4.7.6     | 255.255.255.252 | 16.4.7.4        | 16.4.7.7          | → DistFiber04 |
| <b>SA-Edge-SW20111</b> | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| SA-Edge-SW20111        | F0/0      | 16.4.6.14    | 255.255.255.252 | 16.4.6.12       | 16.4.6.15         | → DistFiber03 |
| SA-Edge-SW20111        | F1/0      | 16.4.7.14    | 255.255.255.252 | 16.4.7.12       | 16.4.7.15         | → DistFiber04 |
| <b>SA-Edge-SW20211</b> | —         | —            | —               | —               | —                 | <b>N/A</b>    |
| SA-Edge-SW20211        | F0/0      | 16.4.6.2     | 255.255.255.252 | 16.4.6.0        | 16.4.6.3          | → DistFiber03 |
| SA-Edge-SW20211        | F1/0      | 16.4.7.2     | 255.255.255.252 | 16.4.7.0        | 16.4.7.3          | → DistFiber04 |

|                        |      |           |                 |           |           |               |
|------------------------|------|-----------|-----------------|-----------|-----------|---------------|
| <b>SA-Edge-SW30111</b> | –    | –         | –               | –         | –         | <b>N/A</b>    |
| SA-Edge-SW30111        | F0/0 | 16.4.6.18 | 255.255.255.252 | 16.4.6.16 | 16.4.6.19 | → DistFiber03 |
| SA-Edge-SW30111        | F1/0 | 16.4.7.18 | 255.255.255.252 | 16.4.7.16 | 16.4.7.19 | → DistFiber04 |
| <b>SA-Edge-SW30211</b> | –    | –         | –               | –         | –         | <b>N/A</b>    |
| SA-Edge-SW30211        | F0/0 | 16.4.6.22 | 255.255.255.252 | 16.4.6.20 | 16.4.6.23 | → DistFiber03 |
| SA-Edge-SW30211        | F1/0 | 16.4.7.22 | 255.255.255.252 | 16.4.7.20 | 16.4.7.23 | → DistFiber04 |

Table 3: Edge Layer Switch Interface IP Addressing

## Server Block 01

| VLAN | Name          | Interface | IPv4 Address | Subnet Mask     | Network Address | Broadcast Address | Gateway     |
|------|---------------|-----------|--------------|-----------------|-----------------|-------------------|-------------|
| 311  | SA-ToR-SWSB01 | Vlan311   | 16.3.63.1    | 255.255.255.192 | 16.3.63.0       | 16.3.63.63        | N/A         |
| 321  | SA-ToR-SWSB01 | Vlan321   | 16.3.63.65   | 255.255.255.192 | 16.3.63.64      | 16.3.63.127       | N/A         |
| 331  | SA-ToR-SWSB01 | Vlan331   | 16.3.63.129  | 255.255.255.192 | 16.3.63.128     | 16.3.63.191       | N/A         |
| 311  | SA-ToR-SWSB02 | Vlan311   | 16.3.63.2    | 255.255.255.192 | 16.3.63.0       | 16.3.63.63        | N/A         |
| 321  | SA-ToR-SWSB02 | Vlan321   | 16.3.63.66   | 255.255.255.192 | 16.3.63.64      | 16.3.63.127       | N/A         |
| 331  | SA-ToR-SWSB02 | Vlan331   | 16.3.63.130  | 255.255.255.192 | 16.3.63.128     | 16.3.63.191       | N/A         |
| 311  | DHCP Server   | E1        | 16.3.63.10   | 255.255.255.192 | 16.3.63.0       | 16.3.63.63        | 16.3.63.1   |
| 321  | NTP Server    | fa0/0     | 16.3.63.74   | 255.255.255.192 | 16.3.63.64      | 16.3.63.127       | 16.3.63.65  |
| 321  | NTP Server    | fa0/1     | 16.3.63.75   | 255.255.255.192 | 16.3.63.64      | 16.3.63.127       | 16.3.63.65  |
| 331  | DNS Server    | E3        | 16.3.63.138  | 255.255.255.192 | 16.3.63.128     | 16.3.63.191       | 16.3.63.129 |
| 311  | SA-PC1SB01    | E1        | 16.3.63.20   | 255.255.255.192 | 16.3.63.0       | 16.3.63.63        | 16.3.63.1   |
| 321  | SA-PC2SB01    | E2        | 16.3.63.84   | 255.255.255.192 | 16.3.63.64      | 16.3.63.127       | 16.3.63.65  |
| 331  | SA-PC3SB01    | E3        | 16.3.63.148  | 255.255.255.192 | 16.3.63.128     | 16.3.63.191       | 16.3.63.129 |

Table 4: Server Block VLAN and Service IP Addressing

## Building 101

| VLAN | Name            | Interface | IPv4 Address | Subnet Mask   | Network Address | Broadcast Address | Gateway   |
|------|-----------------|-----------|--------------|---------------|-----------------|-------------------|-----------|
| 311  | SA-Edge-SW10111 | Vlan311   | 16.1.82.1    | 255.255.255.0 | 16.1.82.0       | 16.1.82.255       | N/A       |
| 321  | SA-Edge-SW10111 | Vlan321   | 16.1.83.1    | 255.255.255.0 | 16.1.83.0       | 16.1.83.255       | N/A       |
| 331  | SA-Edge-SW10111 | Vlan331   | 16.1.84.1    | 255.255.255.0 | 16.1.84.0       | 16.1.84.255       | N/A       |
| 311  | SA-PC1101       | E1        | 16.1.82.10   | 255.255.255.0 | 16.1.82.0       | 16.1.82.255       | 16.1.82.1 |
| 321  | SA-PC2101       | E2        | 16.1.83.10   | 255.255.255.0 | 16.1.83.0       | 16.1.83.255       | 16.1.83.1 |
| 331  | SA-PC3101       | E3        | 16.1.84.10   | 255.255.255.0 | 16.1.84.0       | 16.1.84.255       | 16.1.84.1 |
| 311  | SA-PC4101       | E1        | 16.1.82.11   | 255.255.255.0 | 16.1.82.0       | 16.1.82.255       | 16.1.82.1 |
| 321  | SA-PC5101       | E2        | 16.1.83.11   | 255.255.255.0 | 16.1.83.0       | 16.1.83.255       | 16.1.83.1 |
| 331  | SA-PC6101       | E3        | 16.1.84.11   | 255.255.255.0 | 16.1.84.0       | 16.1.84.255       | 16.1.84.1 |

Table 5.1: Building Access Layer VLAN IP Addressing

## Building 102

| VLAN | Name            | Interface | IPv4 Address | Subnet Mask   | Network Address | Broadcast Address | Gateway   |
|------|-----------------|-----------|--------------|---------------|-----------------|-------------------|-----------|
| 311  | SA-Edge-SW10211 | Vlan311   | 16.1.85.1    | 255.255.255.0 | 16.1.85.0       | 16.1.85.255       | N/A       |
| 321  | SA-Edge-SW10211 | Vlan321   | 16.1.86.1    | 255.255.255.0 | 16.1.86.0       | 16.1.86.255       | N/A       |
| 331  | SA-Edge-SW10211 | Vlan331   | 16.1.87.1    | 255.255.255.0 | 16.1.87.0       | 16.1.87.255       | N/A       |
| 311  | SA-PC1102       | E1        | 16.1.85.10   | 255.255.255.0 | 16.1.85.0       | 16.1.85.255       | 16.1.85.1 |
| 321  | SA-PC2102       | E2        | 16.1.86.10   | 255.255.255.0 | 16.1.86.0       | 16.1.86.255       | 16.1.86.1 |
| 331  | SA-PC3102       | E3        | 16.1.87.10   | 255.255.255.0 | 16.1.87.0       | 16.1.87.255       | 16.1.87.1 |
| 311  | SA-PC4102       | E1        | 16.1.85.11   | 255.255.255.0 | 16.1.85.0       | 16.1.85.255       | 16.1.85.1 |
| 321  | SA-PC5102       | E2        | 16.1.86.11   | 255.255.255.0 | 16.1.86.0       | 16.1.86.255       | 16.1.86.1 |
| 331  | SA-PC6102       | E3        | 16.1.87.11   | 255.255.255.0 | 16.1.87.0       | 16.1.87.255       | 16.1.87.1 |

Table 5.2: Building Access Layer VLAN IP Addressing

## Building 201

| VLAN | Name            | Interface | IPv4 Address | Subnet Mask   | Network Address | Broadcast Address | Gateway   |
|------|-----------------|-----------|--------------|---------------|-----------------|-------------------|-----------|
| 311  | SA-Edge-SW20111 | Vlan311   | 16.1.88.1    | 255.255.255.0 | 16.1.88.0       | 16.1.88.255       | N/A       |
| 321  | SA-Edge-SW20111 | Vlan321   | 16.1.89.1    | 255.255.255.0 | 16.1.89.0       | 16.1.89.255       | N/A       |
| 331  | SA-Edge-SW20111 | Vlan331   | 16.1.90.1    | 255.255.255.0 | 16.1.90.0       | 16.1.90.255       | N/A       |
| 311  | SA-PC1201       | E1        | 16.1.88.10   | 255.255.255.0 | 16.1.88.0       | 16.1.88.255       | 16.1.88.1 |
| 321  | SA-PC2201       | E2        | 16.1.89.10   | 255.255.255.0 | 16.1.89.0       | 16.1.89.255       | 16.1.89.1 |
| 331  | SA-PC3201       | E3        | 16.1.90.10   | 255.255.255.0 | 16.1.90.0       | 16.1.90.255       | 16.1.90.1 |
| 311  | SA-PC4201       | E1        | 16.1.88.11   | 255.255.255.0 | 16.1.88.0       | 16.1.88.255       | 16.1.88.1 |
| 321  | SA-PC5201       | E2        | 16.1.89.11   | 255.255.255.0 | 16.1.89.0       | 16.1.89.255       | 16.1.89.1 |
| 331  | SA-PC6201       | E3        | 16.1.90.11   | 255.255.255.0 | 16.1.90.0       | 16.1.90.255       | 16.1.90.1 |

Table 5.3: Building Access Layer VLAN IP Addressing

## Building 202

| VLAN | Name            | Interface | IPv4 Address | Subnet Mask   | Network Address | Broadcast Address | Gateway   |
|------|-----------------|-----------|--------------|---------------|-----------------|-------------------|-----------|
| 311  | SA-Edge-SW20211 | Vlan311   | 16.3.95.1    | 255.255.255.0 | 16.3.95.0       | 16.3.95.255       | N/A       |
| 321  | SA-Edge-SW20211 | Vlan321   | 16.3.96.1    | 255.255.255.0 | 16.3.96.0       | 16.3.96.255       | N/A       |
| 331  | SA-Edge-SW20211 | Vlan331   | 16.3.97.1    | 255.255.255.0 | 16.3.97.0       | 16.3.97.255       | N/A       |
| 311  | SA-PC1202       | E1        | 16.3.95.10   | 255.255.255.0 | 16.3.95.0       | 16.3.95.255       | 16.3.95.1 |
| 321  | SA-PC2202       | E2        | 16.3.96.10   | 255.255.255.0 | 16.3.96.0       | 16.3.96.255       | 16.3.96.1 |
| 331  | SA-PC3202       | E3        | 16.3.97.10   | 255.255.255.0 | 16.3.97.0       | 16.3.97.255       | 16.3.97.1 |
| 311  | SA-PC4202       | E1        | 16.3.95.11   | 255.255.255.0 | 16.3.95.0       | 16.3.95.255       | 16.3.95.1 |
| 321  | SA-PC5202       | E2        | 16.3.96.11   | 255.255.255.0 | 16.3.96.0       | 16.3.96.255       | 16.3.96.1 |
| 331  | SA-PC6202       | E3        | 16.3.97.11   | 255.255.255.0 | 16.3.97.0       | 16.3.97.255       | 16.3.97.1 |

Table 5.4: Building Access Layer VLAN IP Addressing



## Building 301

| VLAN | Name            | Interface | IPv4 Address | Subnet Mask   | Network Address | Broadcast Address | Gateway    |
|------|-----------------|-----------|--------------|---------------|-----------------|-------------------|------------|
| 311  | SA-Edge-SW30111 | Vlan311   | 16.3.98.1    | 255.255.255.0 | 16.3.98.0       | 16.3.98.255       | N/A        |
| 321  | SA-Edge-SW30111 | Vlan321   | 16.3.99.1    | 255.255.255.0 | 16.3.99.0       | 16.3.99.255       | N/A        |
| 331  | SA-Edge-SW30111 | Vlan331   | 16.3.100.1   | 255.255.255.0 | 16.3.100.0      | 16.3.100.255      | N/A        |
| 311  | SA-PC1301       | E1        | 16.3.98.10   | 255.255.255.0 | 16.3.98.0       | 16.3.98.255       | 16.3.98.1  |
| 321  | SA-PC2301       | E2        | 16.3.99.10   | 255.255.255.0 | 16.3.99.0       | 16.3.99.255       | 16.3.99.1  |
| 331  | SA-PC3301       | E3        | 16.3.100.10  | 255.255.255.0 | 16.3.100.0      | 16.3.100.255      | 16.3.100.1 |
| 311  | SA-PC4301       | E1        | 16.3.98.11   | 255.255.255.0 | 16.3.98.0       | 16.3.98.255       | 16.3.98.1  |
| 321  | SA-PC5301       | E2        | 16.3.99.11   | 255.255.255.0 | 16.3.99.0       | 16.3.99.255       | 16.3.99.1  |
| 331  | SA-PC6301       | E3        | 16.3.100.11  | 255.255.255.0 | 16.3.100.0      | 16.3.100.255      | 16.3.100.1 |

Table 5.5: Building Access Layer VLAN IP Addressing

## Building 302

| VLAN | Name            | Interface | IPv4 Address | Subnet Mask   | Network Address | Broadcast Address | Gateway    |
|------|-----------------|-----------|--------------|---------------|-----------------|-------------------|------------|
| 311  | SA-Edge-SW30211 | Vlan311   | 16.3.101.1   | 255.255.255.0 | 16.3.101.0      | 16.3.101.255      | N/A        |
| 321  | SA-Edge-SW30211 | Vlan321   | 16.3.102.1   | 255.255.255.0 | 16.3.102.0      | 16.3.102.255      | N/A        |
| 331  | SA-Edge-SW30211 | Vlan331   | 16.3.103.1   | 255.255.255.0 | 16.3.103.0      | 16.3.103.255      | N/A        |
| 311  | SA-PC1302       | E1        | 16.3.101.10  | 255.255.255.0 | 16.3.101.0      | 16.3.101.255      | 16.3.101.1 |
| 321  | SA-PC2302       | E2        | 16.3.102.10  | 255.255.255.0 | 16.3.102.0      | 16.3.102.255      | 16.3.102.1 |
| 331  | SA-PC3302       | E3        | 16.3.103.10  | 255.255.255.0 | 16.3.103.0      | 16.3.103.255      | 16.3.103.1 |
| 311  | SA-PC4302       | E1        | 16.3.101.11  | 255.255.255.0 | 16.3.101.0      | 16.3.101.255      | 16.3.101.1 |
| 321  | SA-PC5302       | E2        | 16.3.102.11  | 255.255.255.0 | 16.3.102.0      | 16.3.102.255      | 16.3.102.1 |
| 331  | SA-PC6302       | E3        | 16.3.103.11  | 255.255.255.0 | 16.3.103.0      | 16.3.103.255      | 16.3.103.1 |

Table 5.6: Building Access Layer VLAN IP Addressing